AUTOMATIC FAN CONTROL SYSTEM

Using Passive Infrared (PIR) Motion Sensor

Technical Project Report

Undergraduate Final-Year Engineering Project | Mentored & Supervised

Arduino Uno • PIR Sensor • Relay Module • LCD Display • Axial Fan

1. EXECUTIVE SUMMARY

This report documents an Arduino-based embedded system designed to automate the operation of an electric fan in indoor environments. The system uses a Passive Infrared (PIR) sensor to detect human presence and activates or deactivates the fan accordingly, eliminating the need for manual control.

The project addresses a practical and growing need in Nigeria and similar climates — reducing unnecessary electricity consumption caused by fans left running in unoccupied rooms. By integrating sensor technology, a microcontroller, and a relay-controlled power circuit, the system delivers a low-cost, energy-efficient, and fully automated solution.

I provided technical mentorship and guidance throughout the design, implementation, simulation, and testing phases of this final-year undergraduate project.

2. PROJECT OVERVIEW

Attribute	Detail
Project Title	Automatic Fan Control System Using PIR Sensor
Project Type	Undergraduate Final-Year Engineering Project
Role	Technical Mentor & Supervisor
Domain	Embedded Systems / Home Automation / IoT
Platform	Arduino Uno (ATmega328P)
Programming Language	C++ (Arduino IDE)
Simulation Tool	Proteus 8 Professional

Total Build Cost	₦38,240 (~$24 USD)

3. PROBLEM STATEMENT

In Nigeria, indoor electric fans are commonly left running in unoccupied rooms due to distraction or oversight. This results in:

- Unnecessary and escalating electricity consumption
- Higher utility bills for households and businesses
- Difficulty for physically challenged users (e.g. visually impaired individuals) in manually operating fans
- Lack of a convenient, low-cost automated alternative suitable for the local market

The proposed system addresses all of these challenges by fully automating fan operation based on real-time occupancy detection.

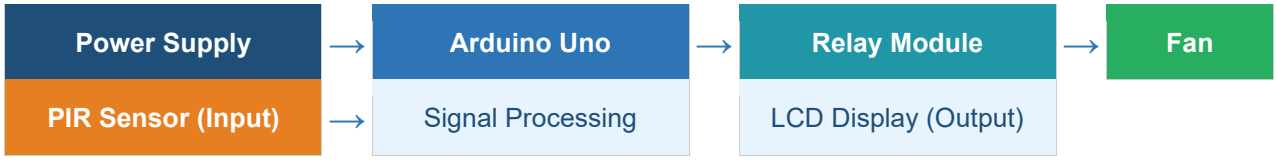
4. OBJECTIVES

1. Automate the operation of an electric fan, replacing manual control entirely.
2. Integrate a PIR motion sensor to detect human presence and trigger the fan accordingly.
3. Display real-time system status (motion detected / fan ON or OFF) on a 16×2 LCD module.
4. Regulate the fan using a relay module controlled by the Arduino microcontroller.
5. Demonstrate the viability of a low-cost, energy-efficient home automation solution.

5. SYSTEM DESIGN & ARCHITECTURE

The system follows a straightforward sense-decide-act architecture. The PIR sensor continuously monitors the room; the Arduino processes the sensor signal and makes a switching decision; the relay then controls the fan's power circuit.

5.1 System Block Diagram



5.2 Operational Logic

The system operates on the following decision loop:

6. The voltage regulator stabilises the 9V input to 5V and powers the Arduino and peripherals.
7. The Arduino continuously reads the PIR sensor output via Analog Pin A0.
8. If the sensor reading exceeds the defined threshold (900 ADC units), motion is confirmed.
9. The Arduino sends a HIGH signal to the relay module (Pin D9), completing the fan's 12V circuit.
10. The LCD is updated to display "Motion: YES" and "Fan: ON".
11. When the PIR reading drops below threshold (no motion), the relay opens, switching the fan off.
12. The LCD updates to "Motion: NO" and "Fan: OFF".

6. HARDWARE COMPONENTS

#	Component	Qty	Cost (₦)	Role in System
1	Arduino Uno (ATmega328P)	1	10,000	Central microcontroller & logic
2	PIR Motion Sensor	1	2,000	Human presence detection
3	5V Relay Module	1	2,200	Switches fan power circuit
4	16×2 LCD with I2C Module	1	9,000	Real-time status display
5	Axial Fan (12V DC)	1	9,000	Air circulation output
6	NPN Transistor (2N2222)	1	400	Fan switching via PWM
7	Voltage Regulator	1	1,080	Stable 5V supply from 9V input

8	9V Battery ×2 + Caps	2	700	Power source
9	Resistors, Breadboard, Wires, Solder	—	3,860	Prototyping & interconnection
	TOTAL		**₦38,240**	**~$24 USD**

7. CIRCUIT CONNECTIONS

The circuit was designed in Proteus 8 and physically assembled on a breadboard before being transferred to a Vero board for permanent soldering. The key wiring connections are summarised below.

Component	Arduino Pin	Notes
PIR Sensor (OUT)	Analog A0	Reads analog motion value; threshold = 900
Relay Module (IN)	Digital D9	HIGH = relay energised = fan ON
LCD I2C (SDA)	Analog A4	I2C data line; address 0x27
LCD I2C (SCL)	Analog A5	I2C clock line
Transistor Base	Digital D9 (via 100Ω)	100Ω resistor limits base current
Voltage Reg OUT	5V pin	Stabilised 5V from 9V input; measured 4.8V
PIR / Relay VCC	5V rail	All logic components share 5V rail
Fan (via Relay NO)	12V supply (external)	Relay COM→12V+; NO→Fan+; Fan−→GND

8. TESTING & RESULTS

Each component was tested individually before system integration. All tests were conducted and verified with hardware connected to a PC running the Arduino IDE Serial Monitor.

Component	Test Objective	Result	Observation
Arduino Uno	Serial output verification	☑ Pass	Message printed every 1s
PIR Sensor	Motion detection (0/1 output)	☑ Pass	1 = motion; 0 = clear
2N2222 Transistor	Fan toggle via PWM pin	☑ Pass	ON/OFF cycle every 2s
Voltage Regulator	Stable 5V output from 9V input	☑ Pass	Measured output: 4.8V
Relay Module	Fan switching on digital signal	☑ Pass	Fan ON/OFF every 2s
16×2 LCD (I2C)	Display test text	☑ Pass	"Hello, World!" displayed
Axial Fan	Powered via relay activation	☑ Pass	Fan spins on relay close

Full System Integration	PIR → Arduino → Relay → Fan cycle	☑ Pass	LCD updates in real time

9. ADVANTAGES & APPLICATIONS

9.1 Key Advantages

- Energy Efficiency: Fan operates only when the room is occupied, reducing electricity waste.
- Full Automation: Eliminates manual switching — no user intervention required after setup.
- User-Friendly Interface: Real-time LCD feedback allows users to monitor system state at a glance.
- Accessibility: Particularly beneficial for users with visual impairments or limited mobility.
- Low Cost: Complete hardware build under ₦40,000 (~$24 USD), well within a consumer budget.
- Scalability: System can be extended to control multiple fans or integrated with HVAC infrastructure.

9.2 Applications

- Residential homes, bedrooms, and living spaces
- Offices and commercial workspaces
- Industrial warehouses and manufacturing facilities
- Smart buildings with broader building automation systems
- Classrooms and educational institutions

10. FUTURE IMPROVEMENTS

Enhancement	Description
Wi-Fi / Bluetooth Control	Add ESP8266 or ESP32 for remote monitoring and control via smartphone app.
Temperature-Based Speed	Integrate DHT11/DHT22 sensor and PWM speed control to vary fan speed with room temperature.
Humidity Sensing	Add humidity sensor for comprehensive environmental monitoring and adaptive control.
Machine Learning	Implement predictive occupancy algorithms based on historical usage patterns to optimise energy further.
Solar Power Integration	Add solar panel and battery management to enable fully off-grid deployment.
HVAC Integration	Extend the system to interface with central heating, ventilation, and air conditioning systems.
Multi-Room Deployment	Network multiple units across rooms using a mesh protocol (e.g. Zigbee or MQTT over Wi-Fi).

11. CONCLUSION

The Automatic Fan Control System successfully demonstrates the practical application of embedded systems technology to a real-world energy efficiency problem. Using an Arduino Uno, a PIR sensor, a relay module, and a 16×2 LCD display, the system reliably detects human presence and automates fan operation without any manual input.

All hardware components were individually tested and validated before integration. The full integrated system performed as designed during testing, with the LCD providing accurate real-time feedback on both motion detection status and fan state.

The project demonstrates strong command of core embedded systems skills: sensor interfacing, microcontroller programming in C++, relay-based power switching, I2C display communication, and low-level circuit design and simulation. Its low bill-of-materials cost (~₦38,240) and clear potential for scalability make it a viable foundation for wider smart-home or IoT deployment.

Mentorship & Supervision Note

My role in this project encompassed end-to-end technical mentorship of the undergraduate student, including:

- Guiding the overall system architecture and component selection decisions
- Reviewing and advising on the Arduino firmware and control logic

- Supporting circuit design, Proteus simulation, and breadboard prototyping phases
- Overseeing the structured testing methodology and verification of results
- Providing feedback on the technical report and project documentation

— End of Report —